

LAB 8

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REG NO: 2023-BSE-022

CLASS: BSE-5A

SUBJECT: CC

LAB TITLE: AWS: Account Setup, IAM, VPC Inventory, EC2, Docker & Gitea

TASK:

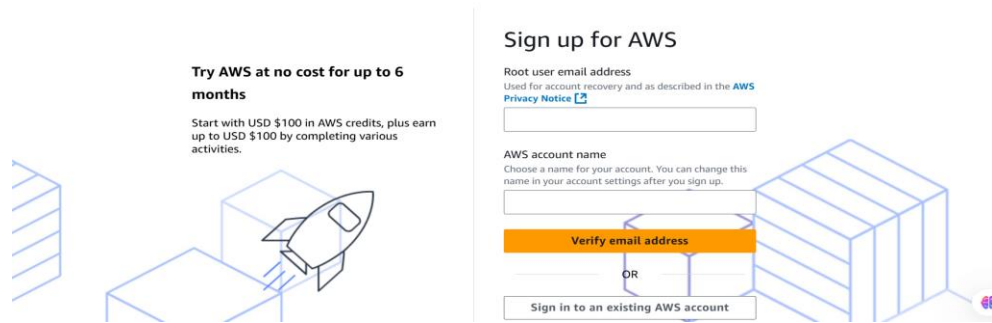
Task 1: Create AWS Root Account & Enable UAE Region

Goal:

Create a new AWS root account and enable the **me-central-1 (UAE)** region.

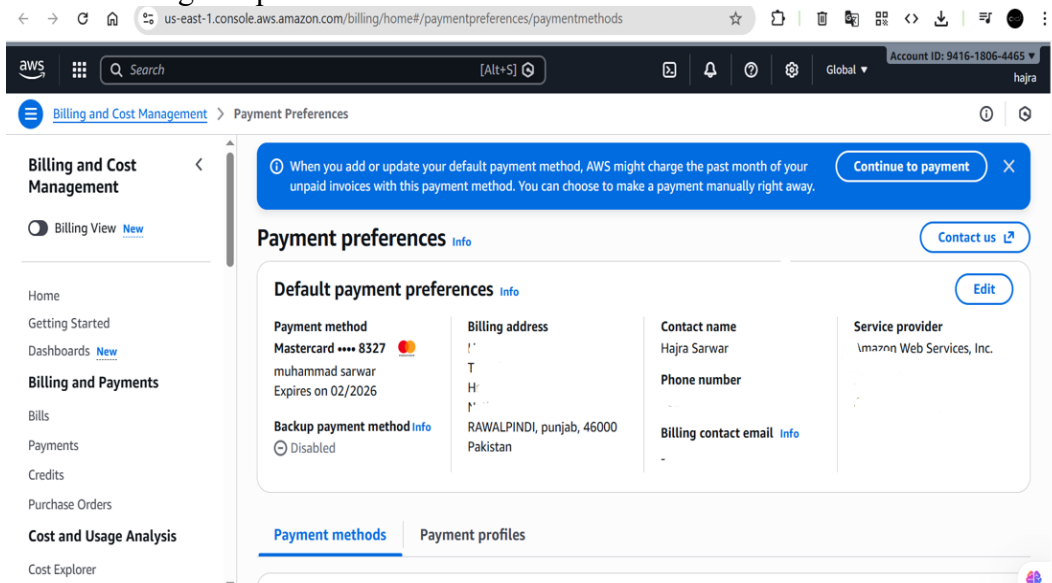
Steps:

1. Opened the browser and navigated to the **AWS Signup page**.

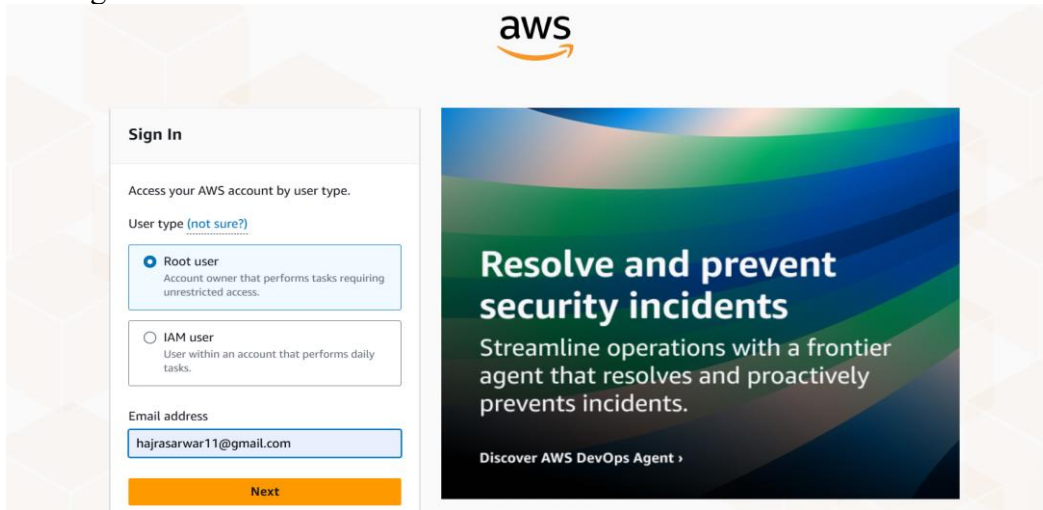


2. Completed registration:

- Account type: Personal
- Plan: AWS Paid Plan
- Entered billing and phone verification details



3. Signed in using the **root email**.

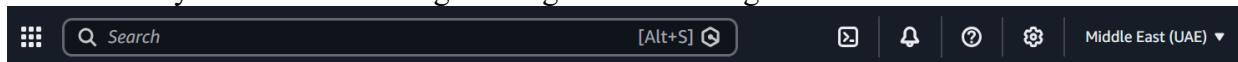


4. Opened the **region selector** from the AWS Console.

5. Enabled and switched the region to **Middle East (UAE) – me-central-1**.

☐	Europe (Spain)	✖ Disabled
☐	Israel (Tel Aviv)	✖ Disabled
☒	Middle East (UAE)	✔ Enabled
☐	Middle East (Bahrain)	✖ Disabled
☐	Mexico (Central)	✖ Disabled

6. Task summary screenshot showing root login and UAE region enabled.



Outcome:

AWS root account was successfully created, and the UAE region was enabled and selected.

Task 2: Create IAM Admin & Lab8User

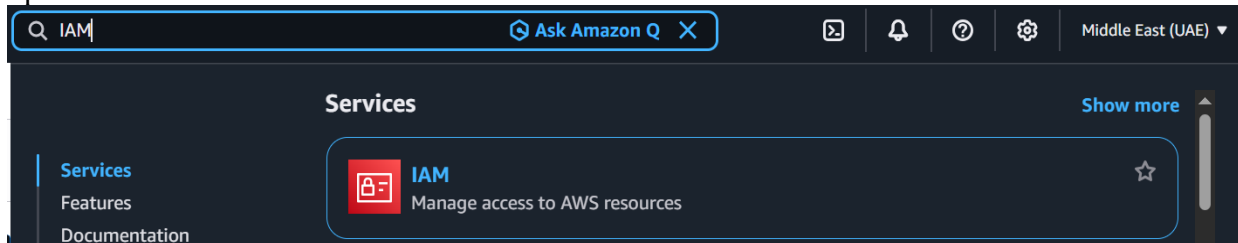
Goal:

Create IAM users with **AdministratorAccess** and verify logins.

Steps:

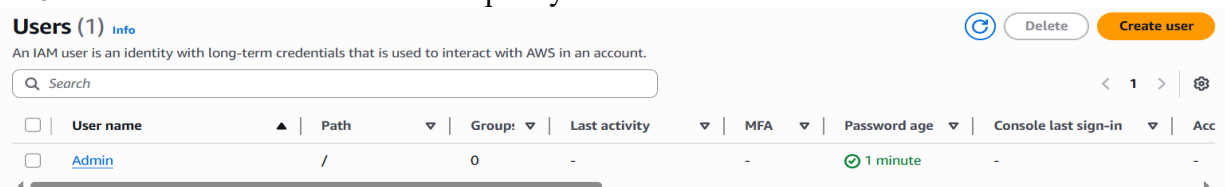
A. Creating Admin User (as Root)

1. Opened **IAM Console** from AWS services.

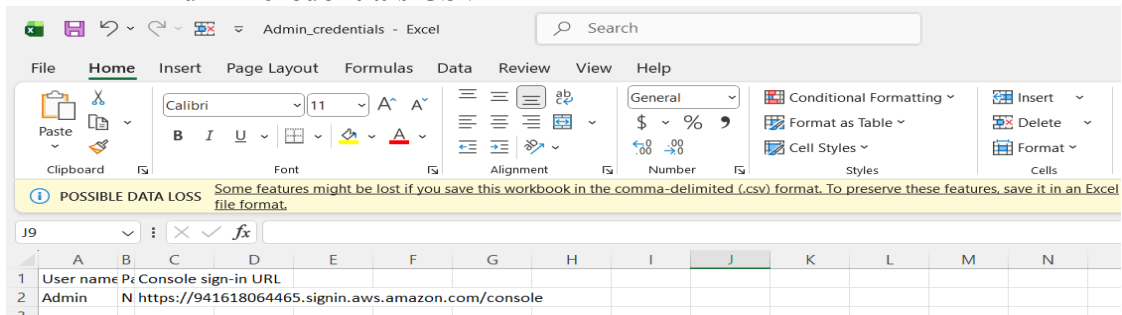


2. Created a new user:

- Username: Admin
- Enabled AWS Management Console access
- Attached AdministratorAccess policy



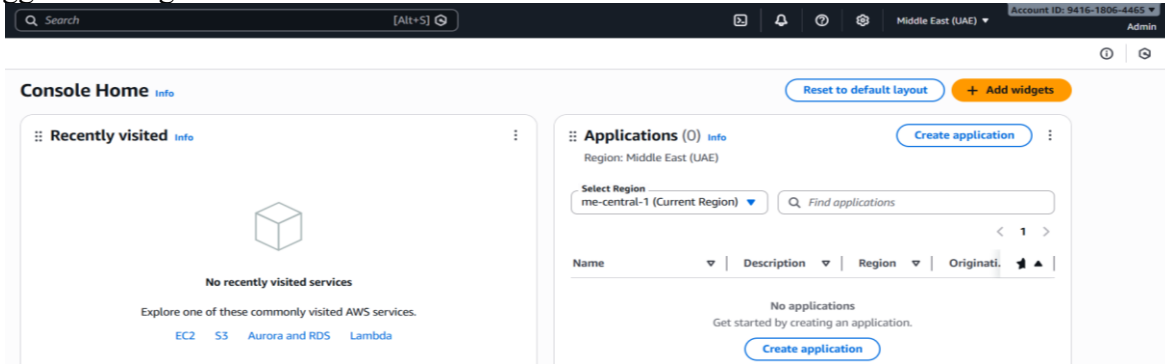
3. Downloaded the **Admin credentials CSV**.



4. Logged out from root account.

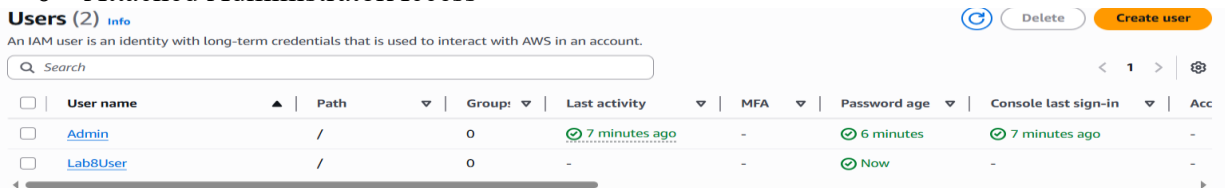
B. Login as Admin & Create Lab8User

5. Logged in using **Admin credentials**.

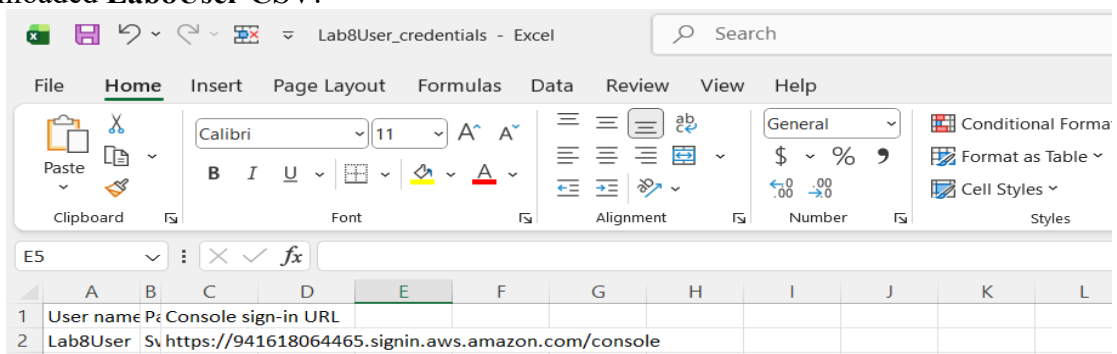


6. Created another IAM user:

- Username: Lab8User
- Console access enabled
- Attached AdministratorAccess



7. Downloaded **Lab8User CSV**.



8. Logged in as **Lab8User**.



9. Task summary screenshot showing both IAM users.

	User name	Path	Group	Last activity	MFA	Password age	Console last sign-in	Acc
☐	Admin	/	0	14 minutes ago	-	13 minutes	14 minutes ago	-
☐	Lab8User	/	0	3 minutes ago	-	2 minutes	3 minutes ago	-

Outcome:

IAM users **Admin** and **Lab8User** were created successfully and verified through login.

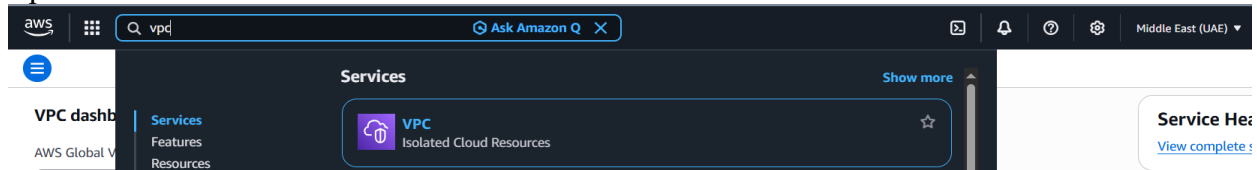
Task 3: Inspect VPC Resources (UAE Region)

Goal:

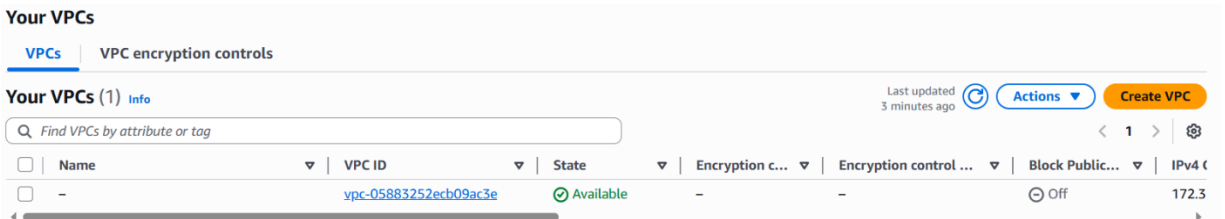
Explore default VPC components in the UAE region.

Steps:

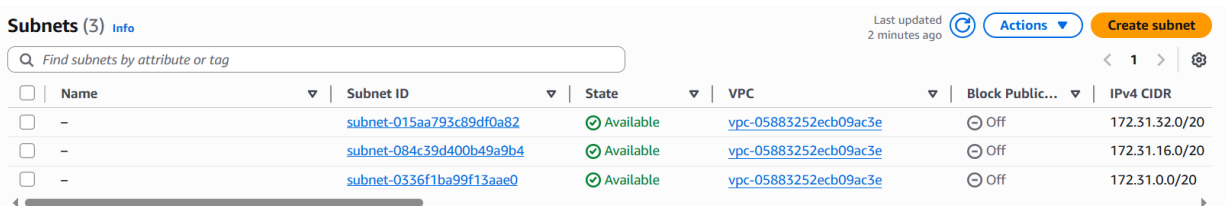
1. Opened VPC Console.



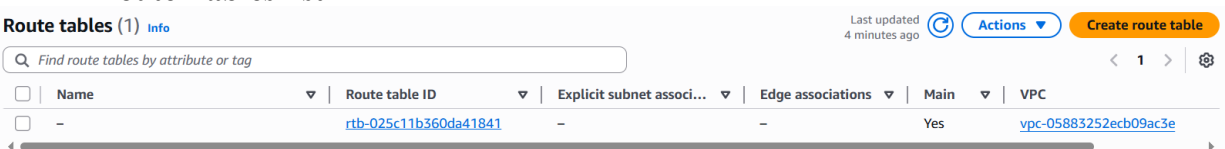
2. Viewed VPCs list.



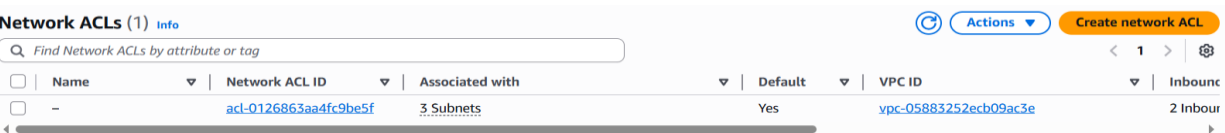
3. Viewed Subnets list.



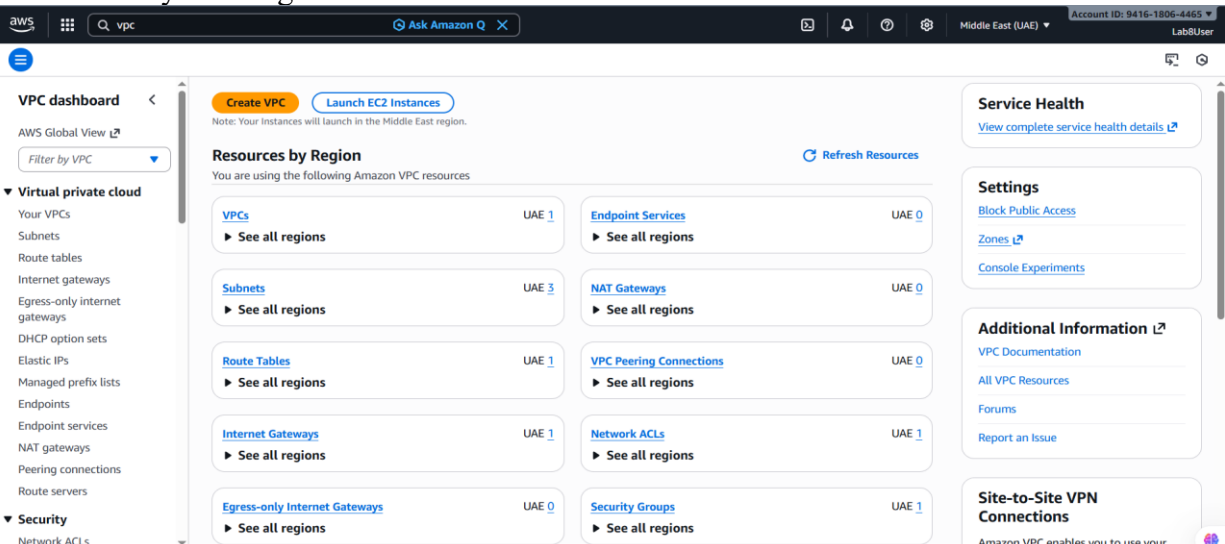
4. Viewed Route Tables list.



5. Viewed Network ACLs list.



6. Task summary showing all VPC resources.



Outcome:

All default VPC resources in the UAE region were successfully inspected.

Task 4: Launch EC2, Install Docker & Deploy Gitea

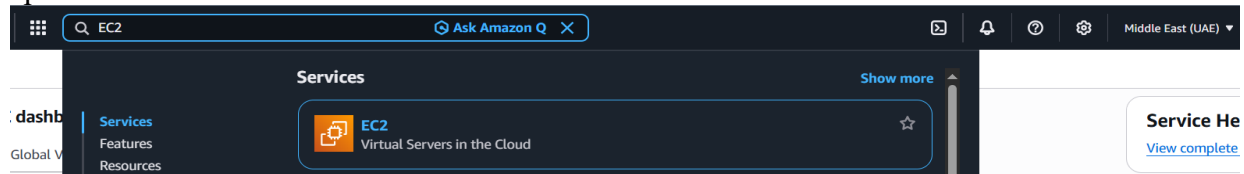
Goal:

Launch an EC2 instance, connect via SSH, install Docker, and deploy **Gitea**.

Steps:

A. Launch EC2 Instance

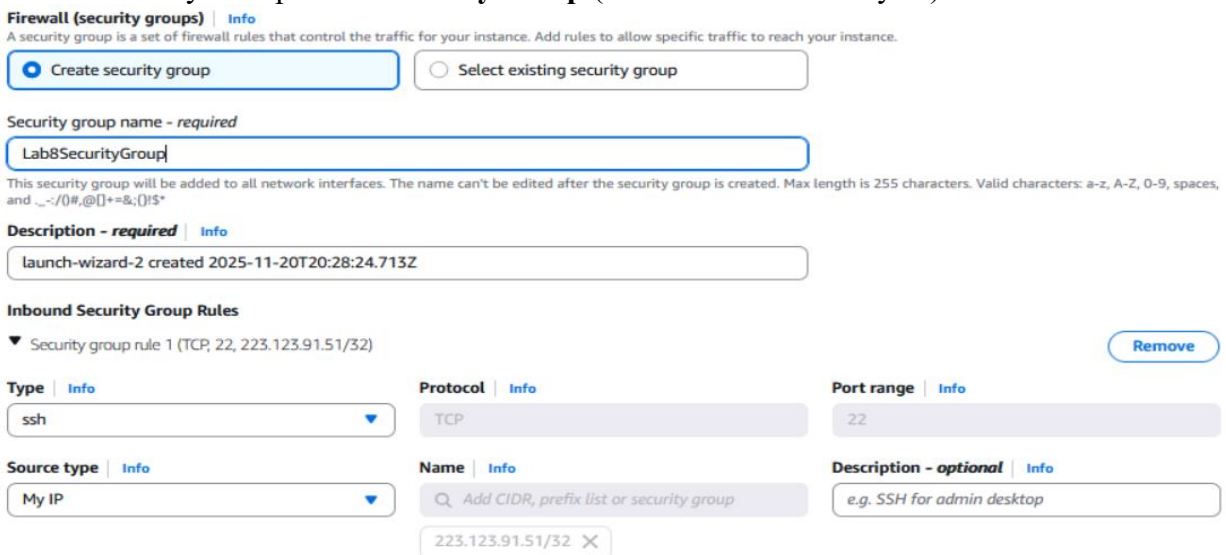
1. Opened EC2 Console.



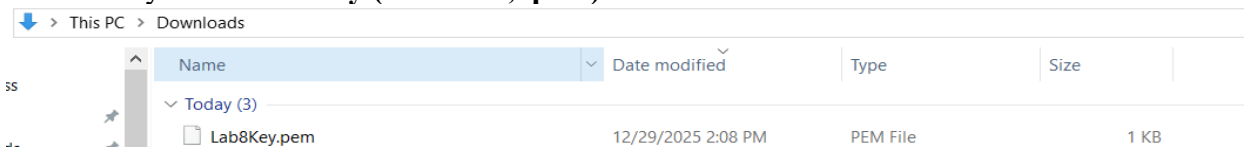
2. Launched a new instance:

- o Name: Lab8Machine
- o AMI: Amazon Linux 2
- o Instance type: t3.micro

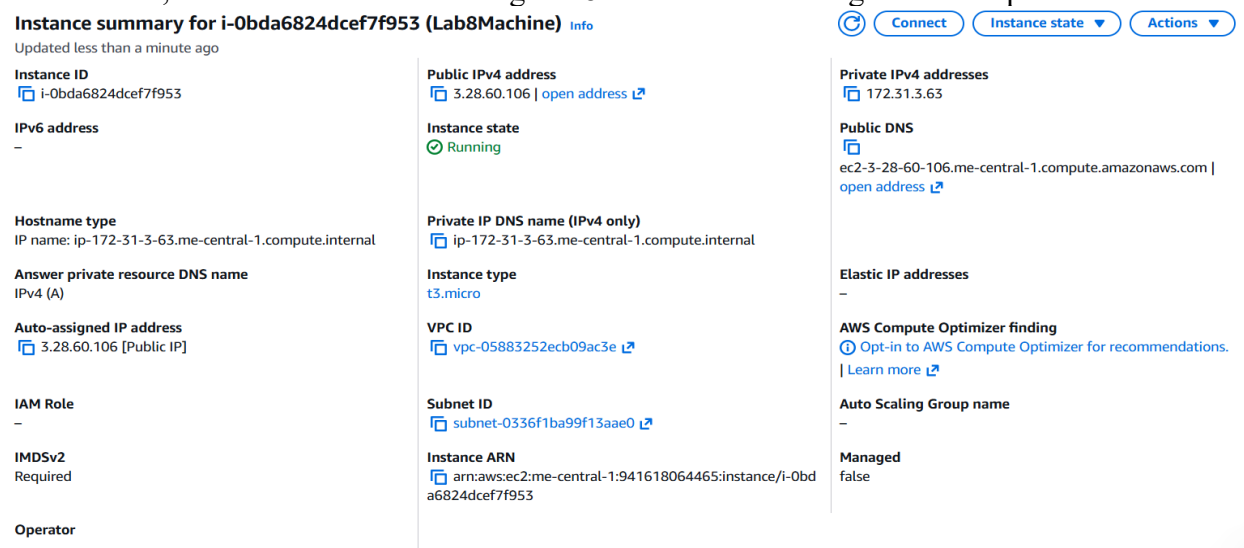
3. Created Security Group: **Lab8SecurityGroup** (SSH allowed from My IP)



4. Created Key Pair: **Lab8Key** (ED25519, .pem) and downloaded it.



5. After launch, EC2 Instances list showing Lab8Machine in "running" state and public IPv4 visible.



B. SSH into EC2

5. Connected from Windows host using:

```
ssh -i Lab8Key.pem ec2-user@<Public-IP>
```

[illegible]

C. Install Docker & Docker Compose

6. Ran installation commands and started Docker service.

```
ec2-user@ip-172-31-3-63 ~]$ sudo yum update -y
Amazon Linux 2023 Kernel livepatch repository
Last metadata expiration check: 0:08:01 ago on Mon Dec 29 10:13:21 2025.
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-3-63 ~]$ sudo yum install -y docker
Last metadata expiration check: 0:37:06 ago on Mon Dec 29 10:13:21 2025.
Dependencies resolved.
```

Package	Architecture	Version	Repository	Size
Installing:				
docker	x86_64	25.0.13-1.amzn2023.0.2	amazonlinux	46 MB
Installing dependencies:				
container-selinux	noarch	4:2.242.0-1.amzn2023	amazonlinux	58 KB
containerd	x86_64	2.1.5-1.amzn2023.0.1	amazonlinux	23 MB
iptables-libse	x86_64	1.8.8-3.amzn2023.0.2	amazonlinux	401 KB
iptables-nft	x86_64	1.8.8-3.amzn2023.0.2	amazonlinux	183 KB
libgroup	x86_64	3.0-1.amzn2023.0.1	amazonlinux	75 KB
libnetfilter_conntrack	x86_64	1.0.8-2.amzn2023.0.2	amazonlinux	58 KB
libnftnl	x86_64	1.0-1-19.amzn2023.0.2	amazonlinux	30 KB
libnftnl	x86_64	1.2.2-2.amzn2023.0.2	amazonlinux	84 KB
pigz	x86_64	2.5-1.amzn2023.0.3	amazonlinux	83 KB
runc	x86_64	1.3.3-2.amzn2023.0.1	amazonlinux	3.9 MB
Transaction Summary				
Install 11 Packages				
Total download size: 74 M				
Installed size: 288 M				
Downloading Packages:				
(1/11): container-selinux-4.2.242.0-1.amzn2023.noarch.rpm			1.4 MB/s 58 kB	00:00
(2/11): iptables-libse-1.8.8-3.amzn2023.0.2.x86_64.rpm			12 MB/s 401 kB	00:00
(3/11): iptables-nft-1.8.8-3.amzn2023.0.2.x86_64.rpm			6.6 MB/s 183 kB	00:00
(4/11): libgroup-3.0-1.amzn2023.0.1.x86_64.rpm			2.3 MB/s 75 kB	00:00
(5/11): libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64.rpm			1.8 MB/s 58 kB	00:00
(6/11): libnftnl-1.0-1-19.amzn2023.0.2.x86_64.rpm			1.0 MB/s 30 kB	00:00
(7/11): libnftnl-1.2.2-2.amzn2023.0.2.x86_64.rpm			2.4 MB/s 84 kB	00:00
(8/11): pigz-2.5-1.amzn2023.0.3.x86_64.rpm			2.6 MB/s 83 kB	00:00
(9/11): containerd-2.1.5-1.amzn2023.0.1.x86_64.rpm			64 MB/s 23 MB	00:00
(10/11): runc-1.3.3-2.amzn2023.0.1.x86_64.rpm			26 MB/s 3.9 MB	00:00
(11/11): docker-25.0.13-1.amzn2023.0.2.x86_64.rpm			67 MB/s 46 MB	00:00
Total			102 MB/s 74 MB	00:00
Running transaction check				
Transaction check succeeded.				
Running transaction test				
Transaction test succeeded.				
Running transaction				
Preparing	:			1/11
Installing	:	runc-1.3.3-2.amzn2023.0.1.x86_64		1/11
Installing	:	containerd-2.1.5-1.amzn2023.0.1.x86_64		2/11
Running scriptlet:	:	containerd-2.1.5-1.amzn2023.0.1.x86_64		2/11
Installing	:	pigz-2.5-1.amzn2023.0.3.x86_64		3/11
Installing	:	libnftnl-1.2.2-2.amzn2023.0.2.x86_64		4/11
Installing	:	libnftnl-1.0-1-19.amzn2023.0.2.x86_64		5/11
Installing	:	libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64		6/11
Installing	:	iptables-libse-1.8.8-3.amzn2023.0.2.x86_64		7/11
Installing	:	iptables-nft-1.8.8-3.amzn2023.0.2.x86_64		8/11
Running scriptlet:	:	iptables-nft-1.8.8-3.amzn2023.0.2.x86_64		8/11
Installing	:	libgroup-3.0-1.amzn2023.0.1.x86_64		9/11
Running scriptlet:	:	container-selinux-4.2.242.0-1.amzn2023.noarch		10/11
Installing	:	container-selinux-4.2.242.0-1.amzn2023.noarch		10/11
Running scriptlet:	:	docker-25.0.13-1.amzn2023.0.2.x86_64		10/11
Installing	:	docker-25.0.13-1.amzn2023.0.2.x86_64		11/11
Running scriptlet:	:	docker-25.0.13-1.amzn2023.0.2.x86_64		11/11
Created symlink /etc/systemd/system/sockets.target.wants/docker.socket → /usr/lib/systemd/system/docker.socket.				
Running scriptlet:	:	container-selinux-4.2.242.0-1.amzn2023.noarch		11/11
Running scriptlet:	:	docker-25.0.13-1.amzn2023.0.2.x86_64		11/11
Verifying	:	container-selinux-4.2.242.0-1.amzn2023.noarch		1/11
Verifying	:	containerd-2.1.5-1.amzn2023.0.1.x86_64		2/11
Verifying	:	docker-25.0.13-1.amzn2023.0.2.x86_64		3/11
Verifying	:	iptables-libse-1.8.8-3.amzn2023.0.2.x86_64		4/11
Verifying	:	iptables-nft-1.8.8-3.amzn2023.0.2.x86_64		5/11
Verifying	:	libgroup-3.0-1.amzn2023.0.1.x86_64		6/11
Verifying	:	libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64		7/11
Verifying	:	libnftnl-1.0-1-19.amzn2023.0.2.x86_64		8/11
Verifying	:	libnftnl-1.2.2-2.amzn2023.0.2.x86_64		9/11
Verifying	:	pigz-2.5-1.amzn2023.0.3.x86_64		10/11
Verifying	:	runc-1.3.3-2.amzn2023.0.1.x86_64		11/11

D. Deploy Gitea

- Created `compose.yaml` and pasted Gitea configuration.

```
ec2-user@ip-172-31-3-63:~  
  container_name gitea  
    environment  
      - DB_TYPE=postgres  
        - DB_HOST=db:5432  
          - DB_NAME=gitea  
            - DB_USER=gitea  
              - DB_PASSWD=gitea  
                restart always  
              volumes  
                - gitea:/data  
              ports  
                - 3000:3000
```


8. Verified `compose.yaml` file exists.

```
[ec2-user@ip-172-31-3-63 ~]$ ls
compose.yaml
```

9. Added user to docker group and reconnected.

```
[ec2-user@ip-172-31-3-63 ~]$ groups
ec2-user adm wheel systemd-journal
[ec2-user@ip-172-31-3-63 ~]$ sudo usermod -aG docker $USER
[ec2-user@ip-172-31-3-63 ~]$ groups
ec2-user adm wheel systemd-journal
[ec2-user@ip-172-31-3-63 ~]$ exit
logout
Connection to 3.28.60.106 closed.

PS C:\Windows\system32> ssh -i "D:\FJWU\Semester 5\Cloud Computing\CC Lab-Engr. Shoaib\Lab8\Lab8Key.pem" ec2-user@3.28.60.106

#
#####
      \   /     Amazon Linux 2023
     _/  _/
    /_  /_
   /_  /_
  /_  /_
 /_  /_
/_  /_

https://aws.amazon.com/linux/amazon-linux-2023

Last login: Mon Dec 29 10:11:52 2025 from 39.60.136.17
[ec2-user@ip-172-31-3-63 ~]$ groups
ec2-user adm wheel systemd-journal docker
```

10. Ran `docker compose up -d`.

```
[ec2-user@ip-172-31-3-63 ~]$ docker compose up -d
WARN[0000] /home/ec2-user/compose.yaml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[*] up 16/18
[*] up 23/23tgres:alpine [*****] 112MB / 112MB Pulling
  Image postgres:alpine Pulled
    1874353ec8d Pulled
    51a9324a2bdc Pulled
    1e827ac0fa1e Pulled
    0629ac8b8cc5 Pulled
    2e50a44abddf Pulled
    4465107e1675 Pulled
    cc2ee20b6816 Pulled
    0df886289dd1 Pulled
    98aeecc8da868 Pulled
    b747867e61cb Pulled
  Image gitea/gitea:latest Pulled
    2d35ebdb57d9 Pulled
    9f4e672c1f34 Pulled
    4da94b3cc809 Pulled
    8e016ece8bd3 Pulled
    346e493cb6cb Pulled
    95215379f1d1 Pulled
  Network ec2-user_webnet Created
  Volume ec2-user_gitea_postgres Created
  Volume ec2-user_gitea Created
  Container gitea_db Created
  Container gitea Created
```

11. Edited Security Group inbound rules to allow port 3000.

Edit inbound rules [Info](#)

Inbound rules control the incoming traffic that's allowed to reach the instance.

Security group rule ID	Type Info	Protocol Info	Port range	Source Info	Description - optional Info
-	Custom TCP	TCP	3000	Anywh...	

0.0.0.0/0 X

Delete

12. Accessed Gitea using browser, run <http://3.28.60.106:3000/> and completed setup.

Install page:

← → ↻ 🔒 Not secure 3.28.60.106:3000

Initial Configuration

If you run Gitea inside Docker, please read the [documentation](#) before changing any settings.

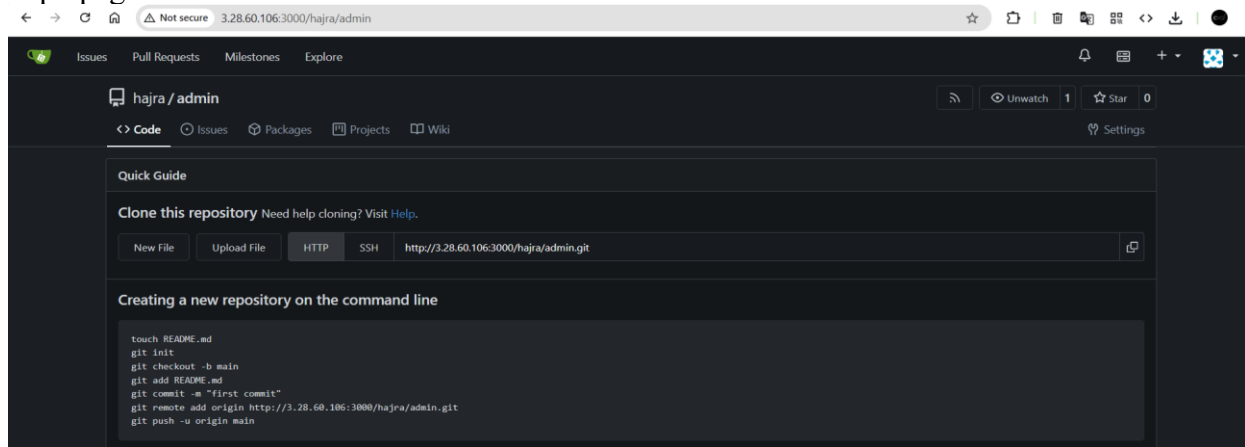
Database Settings

Gitea requires MySQL, PostgreSQL, MSSQL, SQLite3 or TiDB (MySQL protocol).

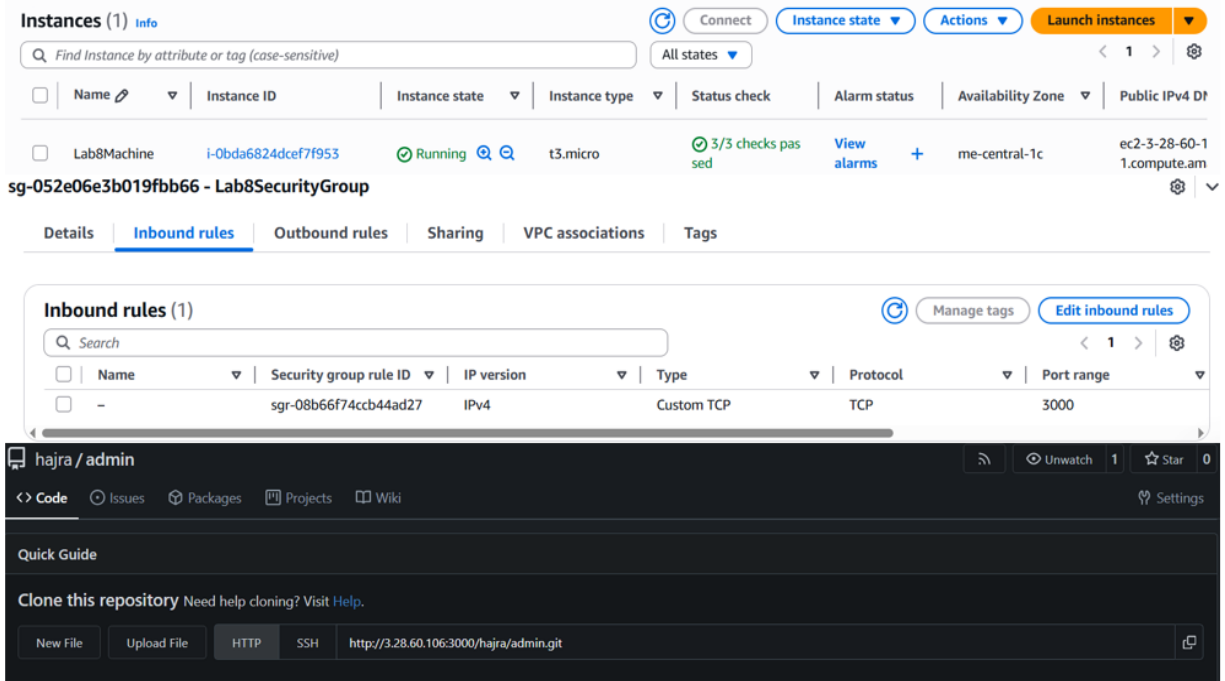
Database Type *	PostgreSQL
Host *	db:5432
Username *	gitea
Password *	*****
Database Name *	gitea
SSL *	Disable
Schema	

Leave blank for database default ("public").

Repo page:



13. Task summary showing EC2, security group, and Gitea UI.



Outcome:

EC2 instance launched, Docker installed, and Gitea deployed successfully.

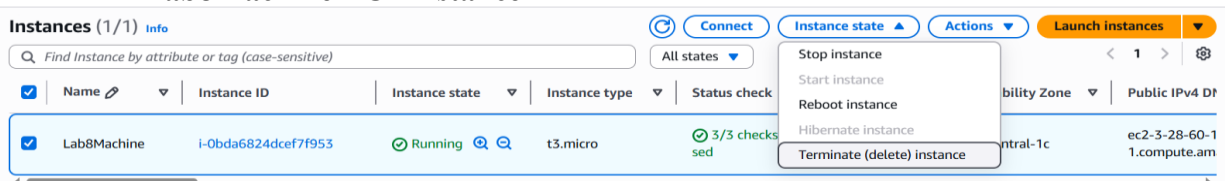
Cleanup: Remove AWS Resources

Goal:

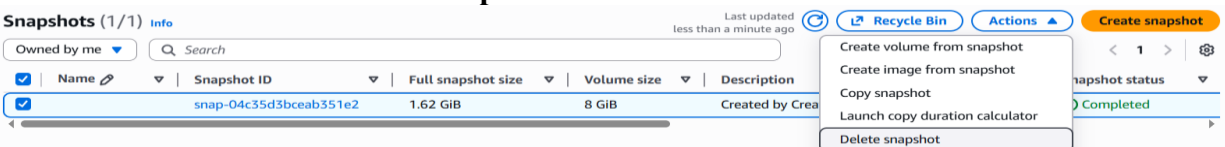
Delete all resources to avoid AWS charges.

Steps:

1. Terminated **Lab8Machine** EC2 instance.



2. Deleted attached **EBS volumes & snapshots**.



3. Deleted Lab8SecurityGroup and Lab8Key.

The screenshot shows the AWS IAM console. At the top, there's a 'Security Groups (1)' section with a table containing one entry: 'sg-0567ae2c936d1bb1c' with a 'default' security group name and 'vpc-05883252ecb09ac3e' VPC ID. Below this is a green success message: 'Successfully deleted 1 key pair'. Underneath is the 'Key pairs (1)' section, which is empty, showing 'No key pairs to display'.

4. Deleted IAM user Lab8User.

The screenshot shows the AWS IAM console with a modal dialog titled 'Delete Lab8User?'. The dialog asks for confirmation to delete the user 'Lab8User' permanently. It shows the user's last activity as '4 hours ago'. A note states: 'Note: Recent activity usually appears within 4 hours. Data is stored for a maximum of 365 days, depending when your region began supporting this feature. Learn more'. Below the note, it says 'To avoid accidental deletions, we ask you to provide additional written consent. To confirm this deletion, type "confirm".' A text input field contains the word 'confirm'. At the bottom are 'Cancel' and 'Delete user' buttons.

5. Verified no active resources remained.

The screenshot shows the AWS Billing console. The 'Summary of balances' section shows 'Total outstanding balance' as '\$0.00 USD' and 'Total available funds' as '\$0.00 USD'. Below this is a 'Payments due (0)' section, which is empty, showing 'No payments due'. A button 'View Transactions' is visible at the bottom.

Outcome:

All AWS resources were safely removed and no active services remained.